

Exam Language Trainer Vision Document

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Introduction

This project develops a web application to help international students studying in their non-native language. The main aim is to improve exam performance by helping students identify vocabulary gaps before exam day. This will be achieved by:

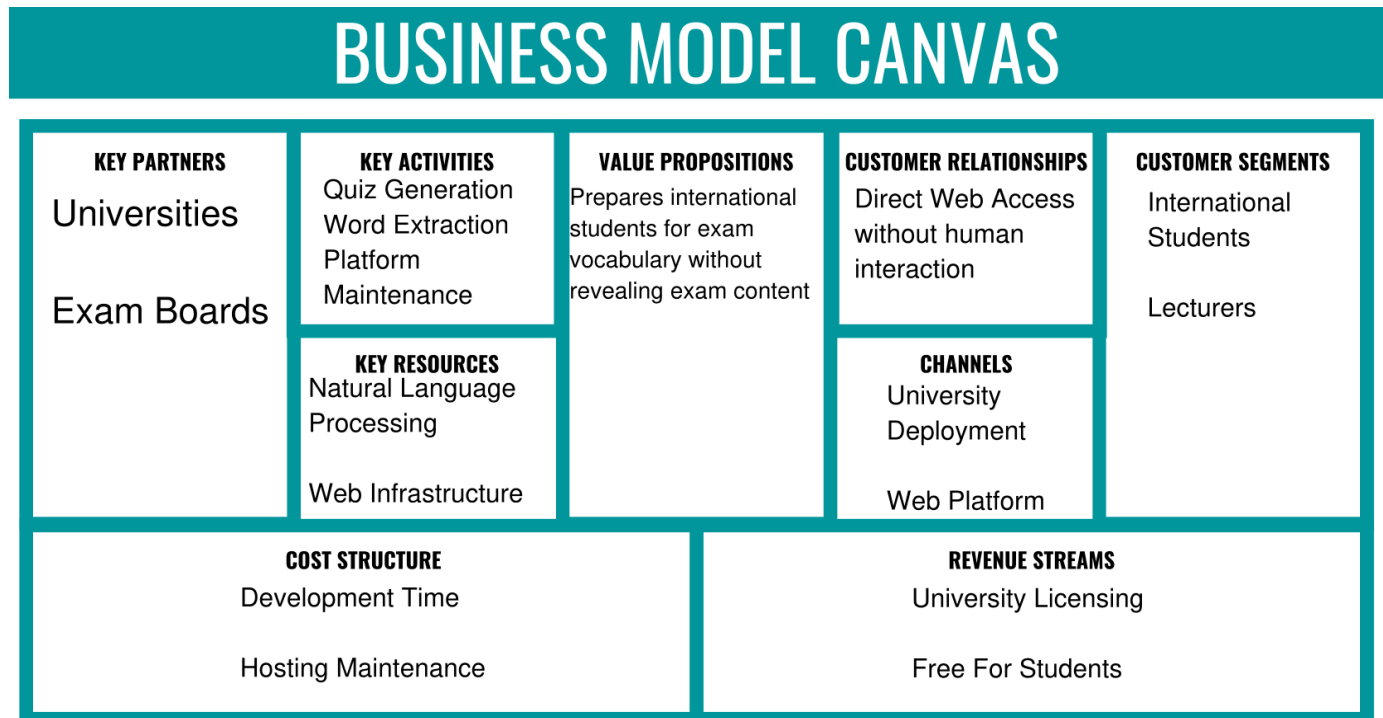
- extracting difficult words from exam papers;
- generating multiple-choice vocabulary quizzes;
- allowing students to test their knowledge without seeing exam content.

The system maintains exam integrity. It avoids extracting domain-specific terms that reveal exam answers and accounts for the student's field of study when assessing word difficulty.

UI Example

The screenshot displays the 'Exam Language Trainer' web application. The top navigation bar is dark blue with a globe icon and the title 'Exam Language Trainer'. It includes links for 'Dashboard', 'My Quizzes', 'Stats', and 'Log Out'. The main content area is light blue and features a welcome message: 'Welcome, Maria! Prepare for Your Upcoming Exam'. Below this, a subtitle reads 'Practice key vocabulary and phrases you might see on the exam.' The interface is divided into three main sections. On the left, the 'Language Prep Quiz - Session 1' section shows the topic 'Economics', '10 Questions', and a 'Time Limit: 15 Minutes', with a 'Start Quiz' button. In the center, the 'Practice Question' section asks 'Which of the following best defines the term "Oligopoly"?' and provides four multiple-choice options: A) A legal agreement to fix prices between companies, B) A market structure with a few dominant firms controlling the market, C) An economic system where all industries are nationalized, and D) A process of reducing tariffs between countries. A 'Submit Answer' button is at the bottom of this section. On the right, the 'Word Review' section lists 'Words to Learn' with four items: 'Ambiguous' (Not clearly defined), 'Fluctuate' (To vary or change irregularly), 'Allocate' (To distribute or assign), and 'Surplus' (An excess or extra amount). Below this, the 'Progress' section shows 'Quiz Completion: 3 / 8 Completed' and 'Vocabulary Mastery: 7 / 10 Words Learned', each with a corresponding progress bar.

Business Model Canvas



Business Case Summary

Universities are increasingly internationalized, with growing numbers of international students studying in non-native languages. Language proficiency represents a significant barrier for international students, affecting both academic performance and later employment.[1]

Current language support focuses on general courses. No tools exist for exam vocabulary preparation that maintain exam integrity.

This project proposes a web application that extracts difficult vocabulary from exam papers and generates practice quizzes. This improves student preparedness while maintaining exam security.

Stakeholders

Lecturers

Lecturers can identify students struggling with vocabulary before exams. This enables early intervention and ensures fairer assessment.

International Students

Students practice difficult vocabulary before exams. This reduces anxiety and improves performance.

Software Overview

The platform will consist of three components:

- **Lecturer Tool (runs in web browser):** Lecturers upload exam papers. The tool processes the document locally on their computer and extracts difficult words. Lecturers review the words and approve quizzes before they are saved to the server.
- **Server:** Stores only the approved quizzes (not the exam papers). Handles student logins and tracks quiz results.
- **Student Interface (web):** Students log in to take vocabulary quizzes. The interface shows their progress and identifies areas where they need more practice.

This design keeps exam papers secure by processing them locally rather than uploading sensitive documents to servers.

Summary of Software Features

- Upload exam papers for local processing
- Extract difficult vocabulary automatically
- Lecturer approval of quiz questions before publication
- Student login and quiz access
- Track student performance and progress
- Secure handling of exam documents

Main Risk Summary

Business

Low adoption by lecturers or students
Competition from existing quiz platforms

Technical Risks

- Browser processing may be slow for large documents
- Defining what makes a word “difficult” is subjective

Project Risks

- Limited time (15 weeks)
- Learning curve for natural language processing

Competitors

Most competitors require uploading documents to their servers. They focus on general quiz generation rather than vocabulary support for international students.

- **Jotform** — AI quiz generation platform
- **AceQuiz AI** — Creates quizzes from uploaded content

This project differs by processing exam papers locally (better security) and targeting vocabulary specifically for non-native speakers.

Work Cited

1. Schueller, Jessica, and Sandra Slotte. “International student retention: going beyond ‘just learn the local language.’” *EAIE*, 23 April 2025, <https://www.eaie.org/resource/international-student-retention-going-beyond-just-learn-the-local-language.html>. Accessed 26 January 2026.

